Early-Exercise and Interest-Rate Calibration:

Least-Squares Monte Carlo for American Options under Black–Scholes and Heston, and Hull–White Calibration to the Swaption Surface

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Abstract

This dissertation develops two pillars of derivatives pricing that lie beyond closed-form European valuation: early-exercise pricing by simulation, and the calibration of a no-arbitrage interest-rate model. For early exercise, the Longstaff–Schwartz least-squares Monte Carlo (LSM) method is implemented and validated against a binomial benchmark in the Black–Scholes world, matching it to within Monte Carlo error across strikes, and is then applied under Heston stochastic-volatility dynamics where no closed form exists, quantifying the early-exercise premium with stochastic volatility. For interest rates, a discount curve is bootstrapped from real U.S. Treasury par yields and the one-factor Hull–White model is fitted exactly to that curve and calibrated to a swaption volatility surface by Jamshidian decomposition, achieving a mean absolute pricing error below four basis points of notional. A synthetic-recovery test recovers the known Hull–White parameters exactly, isolating model error from estimation error. The work demonstrates an end-to-end numerical-methods workflow—simulation regression, lattice benchmarking, curve bootstrapping, and transform-free calibration—across both equity and rates asset classes.

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Chapter 1

Introduction

1.1 Background

Two problems sit at the core of practical derivatives pricing once one leaves the Black–Scholes–Merton European option behind. The first is *early exercise*: American and Bermudan options can be exercised before maturity, which turns pricing into an optimal-stopping problem with no general closed-form solution. The second is *interest-rate calibration*: pricing any rate derivative requires a model that is simultaneously consistent with the observed yield curve and with the prices of liquid volatility instruments such as swaptions.

This dissertation builds and validates solutions to both. For early exercise it uses the least-squares Monte Carlo (LSM) method of Longstaff and Schwartz (2001), which estimates the optimal exercise policy by regressing continuation values on the simulated state. For interest rates it uses the one-factor Hull–White model (Hull and White, 1990), which fits the initial curve exactly and is calibrated to swaptions through the bond-option decomposition of Jamshidian (1989). The two strands share a methodological spine—Monte Carlo, regression, root-finding and non-linear calibration—and together span the equity and rates asset classes.

1.2 Research Questions

1. Does the LSM estimator reproduce a trusted lattice benchmark for American options under Black–Scholes, and what early-exercise premium does it imply?

2. What is the early-exercise premium under Heston stochastic volatility, where no closed-form American price exists?

3. Can a one-factor Hull–White model, fitted exactly to a real Treasury curve, be calibrated to a swaption surface, and how well does a single factor fit?

1.3 Hypothesis

LSM is expected to match the binomial benchmark to within simulation error and to produce an early-exercise premium that grows with moneyness; the Heston American premium should be positive and economically meaningful. The Hull–White model, being a single factor, is expected

to fit the swaption surface well on average but to leave a structured residual in the corners that a two-factor model would remove. The synthetic-recovery test should recover known parameters essentially exactly, locating any market misfit in the model rather than the optimiser.

Chapter 2

Literature Review

2.1 Early-Exercise Pricing

The American option is an optimal-stopping problem (Bensoussan, 1984). Lattice methods (Cox et al., 1979) solve it by backward induction and are the standard low-dimensional benchmark, but they scale poorly with the number of state variables, which rules them out under stochastic volatility. Longstaff and Schwartz (2001) introduced least-squares Monte Carlo, which estimates the continuation value at each potential exercise date by regressing realised discounted future cashflows on basis functions of the current state. Its great advantage is dimensional flexibility: the same regression works whether the underlying is one-dimensional Black–Scholes or a multi-factor stochastic-volatility process, which is precisely why it is used here to price American options under Heston.

2.2 Interest-Rate Models

Short-rate models specify the dynamics of the instantaneous rate. The Vasicek (1977) model is analytically tractable but cannot fit the initial curve; Hull and White (1990) extended it with a time-dependent drift that reproduces any observed curve exactly, making it the workhorse for curve-consistent pricing. Calibrating Hull–White to swaptions relies on Jamshidian (1989), who showed that an option on a coupon bond decomposes into a portfolio of options on zero-coupon bonds once a critical rate is found, giving a near-closed-form swaption price. Brigo and Mercurio (2006) is the standard reference tying the curve bootstrap, the model analytics and the calibration together, and notes the well-understood limitation that a single factor cannot match the full expiry-by-tenor swaption surface—a two-factor or market model is needed for that.

Chapter 3

Data and Methodology

3.1 Data

Two datasets are used. For the equity work, the Heston parameters from a prior calibration to the SPY option surface ($\kappa = 5.25$, $\theta = 0.054$, $\xi = 1.66$, $\rho = -0.64$, $v_0 = 0.022$) are taken as a realistic parameter set. For the rates work, U.S. Treasury par yields for the standard constant-maturity tenors (one month to thirty years), as of early June 2026, are bootstrapped into a discount curve. The swaption volatility surface—a 6×6 grid of at-the-money payer-swaption Black volatilities in expiry and tenor—is synthetic but calibrated to a realistic USD shape (high short-expiry vol decaying in both dimensions), because free swaption data is not publicly available. The methodology is fully real-data-ready: a curve fetch script is provided, and the calibration consumes any surface supplied in the same format.

3.2 Least-Squares Monte Carlo

Consider an American option with payoff $g(S)$ exercisable at dates $t_1 < \cdots < t_M = T$. Working backward from maturity, at each date the holder compares the immediate payoff $g(S_{t_i})$ with the continuation value $\mathbb{E}[\,\text{discounted future cashflow} \mid S_{t_i}]$. LSM estimates that conditional expectation by ordinary least squares: on the in-the-money paths it regresses the realised discounted future cashflow on a polynomial basis of the current underlying price,

$$\widehat{C}(S_{t_i}) = \sum_{k=0}^{d} \beta_k \, S_{t_i}^k, \tag{3.1}$$

and exercises wherever $g(S_{t_i}) > \widehat{C}(S_{t_i})$. The price is the mean of the resulting path cashflows discounted to today. A degree-3 polynomial on in-the-money paths is used throughout, following Longstaff and Schwartz (2001).

3.2.1 Path simulation

Black–Scholes paths are generated as geometric Brownian motion; Heston paths use the full-truncation Euler scheme with antithetic variates. The same LSM routine prices on either path matrix, which is the methodological point.

3.2.2 Benchmark

In the Black–Scholes world the American price is also computed on a Cox–Ross–Rubinstein binomial tree with 2,000 steps, providing an independent benchmark against which LSM is validated. No such benchmark exists under Heston, which is the situation LSM is designed for.

3.3 The Hull–White Model

The short rate follows

$$\mathrm{d}r_t = \big(\theta(t) - a\, r_t\big)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t, \tag{3.2}$$

where the deterministic drift $\theta(t)$ is chosen so that the model reproduces the initial discount curve $P(0,T)$ exactly. Zero-coupon bond prices are affine,

$$P(t,T) = A(t,T)\,e^{-B(t,T)r_t}, \qquad B(t,T) = \frac{1 - e^{-a(T-t)}}{a}, \tag{3.3}$$

with $A(t,T)$ pinned to the initial curve and its instantaneous forward rate.

3.3.1 Curve bootstrap

The discount curve is bootstrapped from par yields by sequential stripping: at each coupon date the par-bond identity $1 = \sum_j c\,P(0,t_j) + (1+c)P(0,T)$ is solved for the next discount factor, interpolating log-linearly in discount factors between nodes. The instantaneous forward rate, needed for $A(t,T)$, is obtained by finite differencing the log discount curve.

3.3.2 Swaption pricing by Jamshidian decomposition

A payer swaption is an option to enter a swap, equivalently a put on a coupon bond struck at par. Following Jamshidian (1989), the critical short rate r^* at which the underlying coupon bond equals par is found by root-finding; because bond prices are monotone in r, the coupon-bond option then decomposes into a sum of options on the constituent zero-coupon bonds, each struck at its value at r^*. Each zero-bond option has a closed-form Hull–White price, so the swaption price is a weighted sum of these.

3.4 Calibration

The two Hull–White parameters (a,σ) are chosen to minimise the squared difference between model swaption prices and the market prices implied by the quoted Black volatilities,

$$\min_{a,\sigma} \sum_i \big(\mathrm{HW}_i(a,\sigma) - \mathrm{Black}_i\big)^2, \tag{3.4}$$

using a multi-start Nelder–Mead search to avoid local minima. The model is fitted exactly to the curve first, so calibration adjusts only the volatility dynamics.

3.5 Validation Design

Each strand is validated independently of market data. LSM is checked against the binomial benchmark and against the theorem that an American call on a non-dividend stock equals its European counterpart. Hull–White calibration is checked by generating a swaption set from known (a, σ) and recovering them from scratch, which separates optimiser error from model misfit.

Chapter 4

Empirical Results

4.1 LSM Validation under Black–Scholes

Table 4.1 compares LSM American-put prices with the binomial benchmark across strikes. The two agree to within Monte Carlo error at every strike, and the early-exercise premium over the European put rises monotonically with moneyness, from 0.04 deep out-of-the-money to 2.74 deep in-the-money—as expected, since early exercise is valuable precisely when the option is deep in-the-money and the time value is small. As a further check, an American call on a non-dividend stock priced by LSM equals its European value, recovering the standard no-early-exercise result.

Figure 4.1 shows the same comparison graphically: the American and European curves coincide far out-of-the-money and diverge as the option moves in-the-money, with the LSM points sitting on the binomial American curve. Figure 4.2 plots the estimated early-exercise boundary—the critical stock price below which immediate exercise is optimal—which rises toward the strike as maturity approaches, the canonical shape for an American put.

4.2 Early Exercise under Heston

Table 4.2 prices the American put under Heston dynamics, where the European leg is computed by the Fourier-cosine method and the American leg by LSM on simulated Heston paths. The early-exercise premium is positive throughout and grows with moneyness, from 0.11 at $K = 90$ to 0.74 at $K = 110$. Because no closed-form American price exists under stochastic volatility, LSM is the practical valuation route, and the consistency of its European-leg behaviour with the COS price (which *is* closed-form) lends confidence to the American figures.

4.3 The Treasury Curve

Figure 4.3 shows the bootstrapped discount curve. The par yields are mapped to a smooth zero-rate curve and the implied instantaneous-forward curve, which sits above the zero curve where the curve slopes upward and below it where it inverts—the correct qualitative relationship. The curve is mildly humped, with a dip around the 2–3 year tenor and a rising long end, consistent with the input par yields.

Table 4.1: LSM American-put prices versus the binomial benchmark (Black–Scholes world).

Strike	LSM	Binomial	European	Early-ex. premium
80	0.729	0.723	0.687	0.036
90	2.468	2.473	2.310	0.162
100	6.073	6.090	5.574	0.516
110	11.950	11.973	10.675	1.298
120	20.090	20.136	17.395	2.741

Notes: $S_0 = 100$, $r = 5\%$, $\sigma = 20\%$, $T = 1$. *LSM uses 200,000 antithetic paths, 50 exercise dates and a degree-3 polynomial basis; the binomial benchmark uses 2,000 steps. LSM matches the benchmark to within Monte Carlo error throughout, and the early-exercise premium over the European put grows with moneyness.*

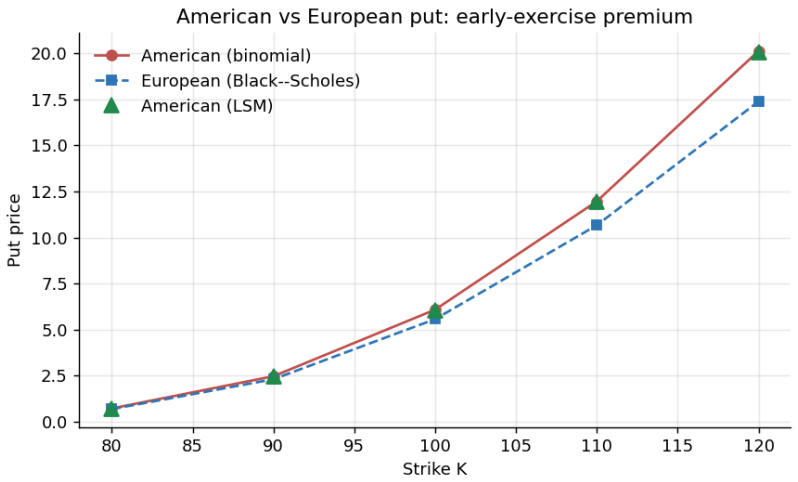


Figure 4.1: American versus European put prices across strikes.

Notes: $S_0 = 100$, $r = 5\%$, $\sigma = 20\%$, $T = 1$. *LSM (green markers) sits on the binomial American curve (red); the European curve (blue) lies below, the gap being the early-exercise premium.*

4.4 Hull–White Calibration

Table 4.3 reports the calibration. The fitted mean reversion is $a = 0.020$ and volatility $\sigma = 0.0131$, with a mean absolute pricing error below four basis points of notional across the 36 swaptions. Figure 4.4 shows the market vol surface and the resulting Hull–White price errors. The fit is good across most of the grid—errors within roughly ten basis points—but a structured residual appears in the long-expiry, mid-tenor corner, where the single factor cannot simultaneously match the level and the curvature of the surface. This is the well-documented limitation of one-factor models (Brigo and Mercurio, 2006); the mean-reversion parameter reaching the lower bound of its search range is a symptom of the same tension. The synthetic-recovery test in Table 4.3 recovers the known parameters exactly, confirming that the residual is genuine model misfit rather than an optimisation artefact.

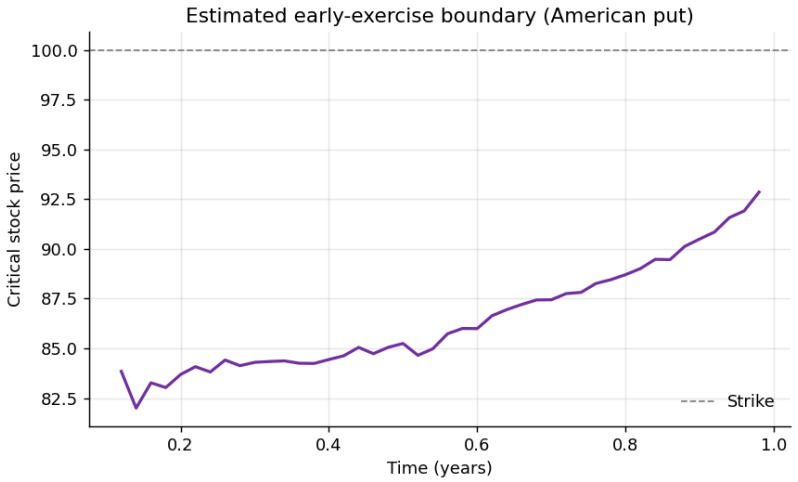


Figure 4.2: Estimated early-exercise boundary for the American put.

Notes: The critical stock price rises toward the strike as expiry approaches, the standard shape for an American put: the continuation region shrinks as time value decays.

Table 4.2: American versus European put under Heston dynamics.

Strike	American (LSM)	European (COS)	Early-ex. premium
90	3.175	3.062	0.113
100	5.669	5.442	0.227
110	10.253	9.513	0.740

Notes: Heston parameters $\kappa = 5.25$, $\theta = 0.054$, $\xi = 1.66$, $\rho = -0.64$, $v_0 = 0.022$ (from the equity calibration), $S_0 = 100$, $r = 5\%$, $T = 1$. The European leg is priced by the COS method; the American leg by LSM on simulated Heston paths. No closed form exists for the American Heston price, so LSM is the practical route.

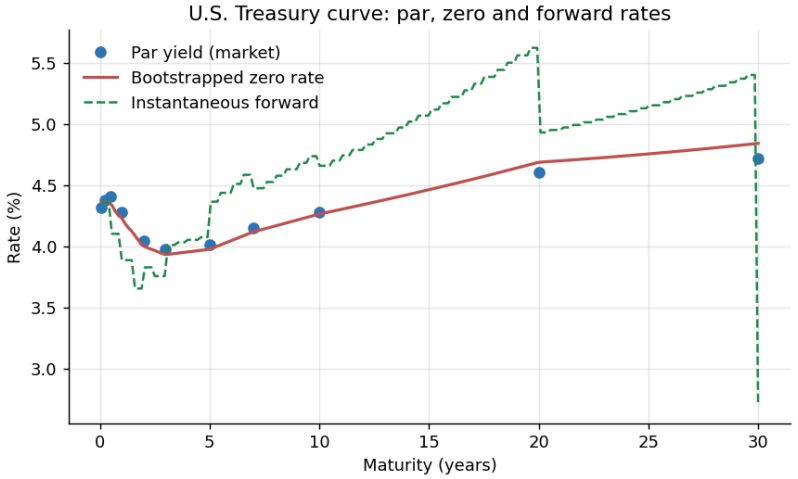


Figure 4.3: U.S. Treasury curve: par, zero and instantaneous-forward rates.

Notes: Par yields (markers) are bootstrapped into the zero curve (red) and differentiated into the instantaneous-forward curve (green dashed). The forward sits above the zero where the curve rises and below where it falls.

Table 4.3: Hull–White calibration and synthetic-recovery test.

	Mean reversion a	Volatility σ
Calibrated to market surface	0.0200	0.01308
Synthetic: true	0.0800	0.01200
Synthetic: recovered	0.0800	0.01200

Notes: Hull–White (a, σ) calibrated to 36 ATM payer swaptions by minimising squared price error, with the model fitted exactly to the bootstrapped Treasury curve. Mean reversion reaches its lower search bound, reflecting a single factor's difficulty in matching the full expiry/tenor structure. The synthetic test recovers known parameters exactly, confirming the optimiser.

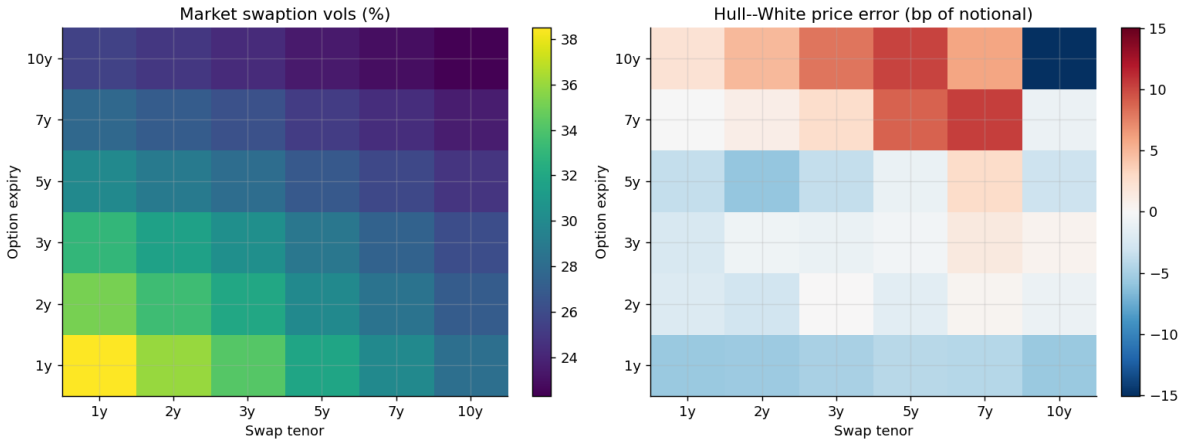


Figure 4.4: Swaption vol surface and Hull–White calibration errors.

Notes: Left: market ATM payer-swaption Black volatilities by expiry and tenor. Right: Hull–White price error in basis points of notional after calibration. Errors are small across most of the surface, with a structured residual in the long-expiry/mid-tenor corner characteristic of a single-factor model.

Chapter 5

Conclusion

5.1 Summary of Findings

This dissertation built and validated solutions to two problems beyond closed-form European pricing. Least-squares Monte Carlo reproduced the binomial American-put benchmark to within simulation error and quantified an early-exercise premium that grows with moneyness; applied under Heston, where no closed form exists, it produced positive and sensible premia. The Hull–White model, fitted exactly to a real Treasury curve and calibrated to a swaption surface by Jamshidian decomposition, priced the surface to within a few basis points, with a structured residual exactly where single-factor theory predicts. A synthetic-recovery test recovered the known rate parameters exactly.

5.2 Contribution

The work demonstrates a complete numerical-methods toolkit across two asset classes: regression-based optimal stopping, lattice benchmarking, antithetic Monte Carlo, curve bootstrapping, affine bond analytics, Jamshidian decomposition, and multi-start non-linear calibration. The validation discipline—benchmarking LSM against a lattice, and Hull–White against a synthetic recovery—is the methodological core, since it establishes that each reported result is trustworthy and locates residual error in the model rather than the method.

5.3 Limitations and Future Research

Several limitations apply. The swaption surface is synthetic; a real surface, though hard to source freely, would make the rates calibration fully empirical. The single-factor Hull–White model cannot fit the whole swaption surface, as the residuals show; a two-factor (G2++) model or a market model would be the natural extension. The LSM estimator carries a known low bias from in-sample regression; a separate out-of-sample pass for the exercise decision would remove it. The Heston American prices use a fixed parameter set; jointly stressing volatility parameters and the early-exercise premium would be informative. Finally, the Treasury curve is a single snapshot, so the calibration is cross-sectional.

5.4 Concluding Remarks

The unifying lesson is that once closed forms run out—at early exercise, and at a full volatility surface—the right response is a validated numerical method paired with an honest benchmark. LSM against a lattice, and Hull–White against a synthetic recovery, each turn an otherwise unverifiable number into a trustworthy one, and each reveals exactly where the method's domain ends: LSM extends cleanly to stochastic volatility, while a single-factor rate model meets its limit at the corners of the swaption surface.

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